BETS

Institute of Technology

Network Design and VLSM Implementation

Assignment 04

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Subject	Network Design & Subnetting

1. Introduction

In this assignment, I have designed a small company network using **Cisco Packet Tracer**. The company has five departments, each needing a different number of IP addresses. To avoid wasting addresses, I used **VLSM (Variable Length Subnet Masking)**, which lets me give each department exactly the size of subnet it needs.

Three routers are used to connect all departments. Two of the routes between routers are configured as **static routes**, and the rest of the network uses **RIP version 2** as the dynamic routing protocol. By the end, all devices can ping each other across the entire network.

2. Network Design

2.1 Departments and Host Requirements

The company has five departments. The table below shows the name of each department, how many hosts it needs, and which router it is connected to.

Department	Hosts Needed	Connected to Router
IT	50	R1
HR	25	R1
Finance	20	R2
Sales	10	R2
Management	5	R3

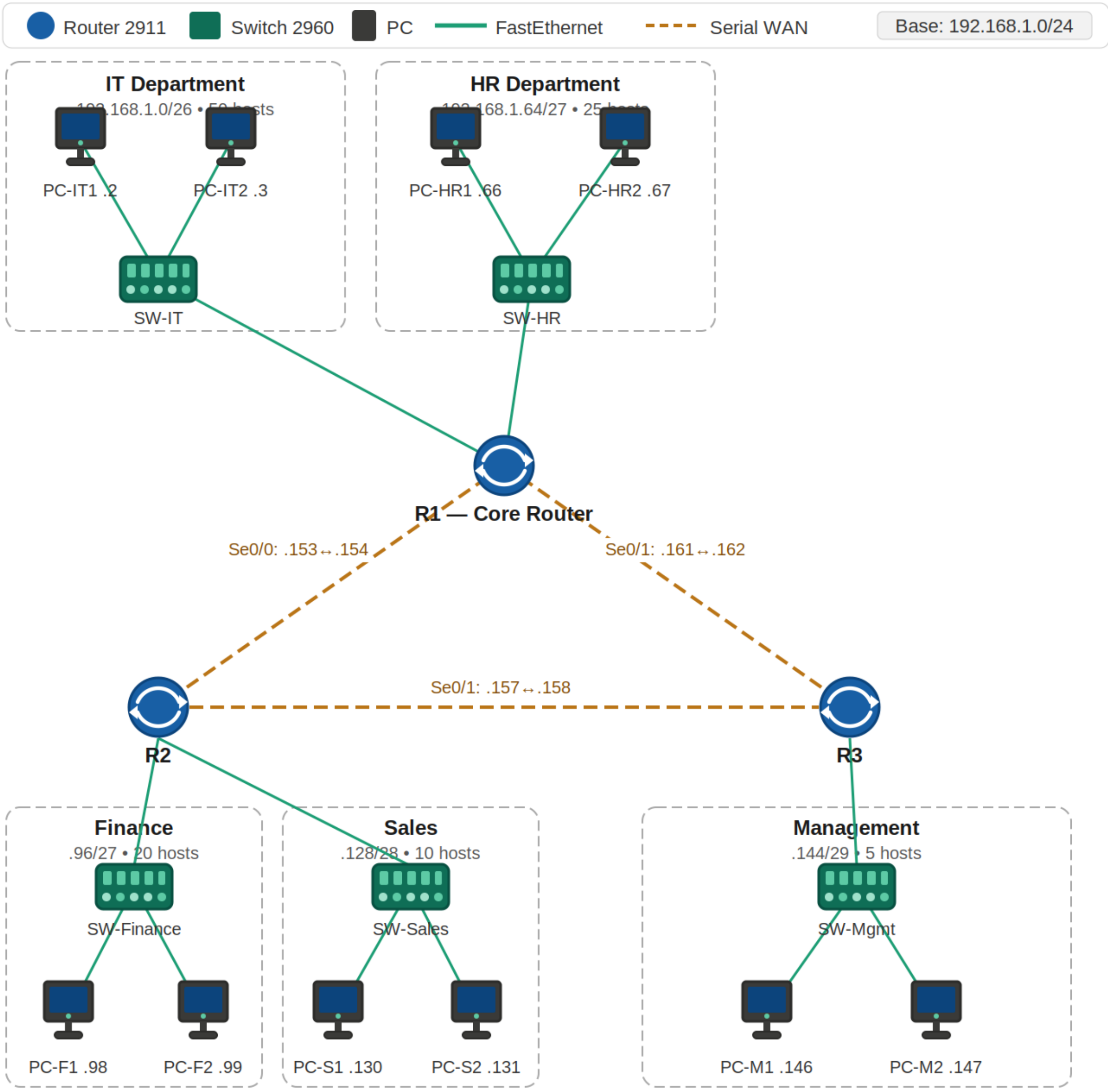
2.2 Base Network Address

The base network address given for this assignment is **192.168.1.0 / 24**. This gives 256 addresses in total (from .0 to .255). All subnets will be carved out of this single /24 block using VLSM.

2.3 Topology Overview

Router R1 is the core router. It connects directly to R2 and R3 via serial WAN links. Each router has one or more LAN interfaces connecting to switches, and each switch connects to the PCs of that department.

Network Topology Diagram — Packet Tracer (Router 2911 / Switch 2960 / PC)



Device	Model	Role
R1	Cisco 2911	Core router — connects IT, HR departments and WAN links to R2 and R3
R2	Cisco 2911	Connects Finance and Sales departments; WAN to R1 and R3
R3	Cisco 2911	Connects Management department; WAN to R1 and R2
SW-IT / SW-HR / SW-Finance / SW-Sales / SW-Mgmt	Cisco 2960	Layer-2 switch per department

PC-IT1, PC-IT2 … PC-M2	Generic PC	Two end devices per department (10 PCs total)

3. VLSM Calculations — Step by Step

3.1 What is VLSM?

VLSM stands for Variable Length Subnet Masking. Instead of splitting a network into equal-sized pieces, VLSM lets you choose a different prefix length for each subnet depending on how many hosts it needs. This stops you from wasting addresses.

The key formula is: **Usable hosts** $= 2^n - 2$ where n is the number of host bits. You subtract 2 because the network address and broadcast address cannot be assigned to hosts.

3.2 Step 1 — Sort Departments Largest to Smallest

With VLSM you must always start with the department that needs the most hosts. If you start with a small subnet you might not leave enough contiguous space for the larger ones. Below the departments are sorted from largest to smallest, including the three point-to-point WAN links between routers which each only need 2 usable addresses.

Order	Subnet Needed For	Hosts Required	Min. Block Size (2^n)	Host Bits (n)	Prefix Length
1	IT Department	50	64 (2^6)	6	/26
2	HR Department	25	32 (2^5)	5	/27
3	Finance Department	20	32 (2^5)	5	/27
4	Sales Department	10	16 (2^4)	4	/28
5	Management Department	5	8 (2^3)	3	/29
6	WAN R1 — R2	2	4 (2^2)	2	/30
7	WAN R2 — R3	2	4 (2^2)	2	/30
8	WAN R1 — R3	2	4 (2^2)	2	/30

For each row I calculated the smallest power of 2 that is greater than the number of hosts required (remembering to add 2 for the network and broadcast). For example, IT needs 50 hosts so I need at least 52 addresses, and the next power of 2 above 52 is $64 = 2^6$, which gives a /26 prefix.

3.3 Step 2 — Allocate Subnets One by One

Starting from 192.168.1.0, I allocate subnets in order from largest to smallest. After each allocation the next subnet starts right after the broadcast address of the previous one.

Subnet 1 — IT Department

Hosts needed: 50 → using /26 (gives $2^6 - 2 = 62$ usable addresses)

Starting from 192.168.1.0, the block size for a /26 is $256 - 192 = 64$.

Field	Value
Network address	192.168.1.0
Subnet mask	255.255.255.192 (/26)
First usable host	192.168.1.1
Last usable host	192.168.1.62
Broadcast address	192.168.1.63
Usable hosts	62 (50 used, 12 spare)
Next subnet starts	192.168.1.64

Subnet 2 — HR Department

Hosts needed: 25 → using /27 (gives $2^5 - 2 = 30$ usable addresses)

Starting from 192.168.1.64. Block size = $256 - 224 = 32$.

Field	Value
Network address	192.168.1.64
Subnet mask	255.255.255.224 (/27)
First usable host	192.168.1.65
Last usable host	192.168.1.94
Broadcast address	192.168.1.95
Usable hosts	30 (25 used, 5 spare)
Next subnet starts	192.168.1.96

Subnet 3 — Finance Department

Hosts needed: 20 → using /27 (gives 30 usable addresses)

Starting from 192.168.1.96. Block size = 32.

Field	Value
Network address	192.168.1.96
Subnet mask	255.255.255.224 (/27)
First usable host	192.168.1.97
Last usable host	192.168.1.126
Broadcast address	192.168.1.127
Usable hosts	30 (20 used, 10 spare)
Next subnet starts	192.168.1.128

Subnet 4 — Sales Department

Hosts needed: 10 → using /28 (gives $2^4 - 2 = 14$ usable addresses)

Starting from 192.168.1.128. Block size = 256 − 240 = 16.

Field	Value
Network address	192.168.1.128
Subnet mask	255.255.255.240 (/28)
First usable host	192.168.1.129
Last usable host	192.168.1.142
Broadcast address	192.168.1.143
Usable hosts	14 (10 used, 4 spare)
Next subnet starts	192.168.1.144

Subnet 5 — Management Department

Hosts needed: 5 → using /29 (gives $2^3 - 2 = 6$ usable addresses)

Starting from 192.168.1.144. Block size = 256 − 248 = 8.

Field	Value
Network address	192.168.1.144
Subnet mask	255.255.255.248 (/29)
First usable host	192.168.1.145
Last usable host	192.168.1.150
Broadcast address	192.168.1.151
Usable hosts	6 (5 used, 1 spare)
Next subnet starts	192.168.1.152

WAN Links — Serial Point-to-Point (/30)

Each serial link between two routers only needs two usable addresses, one for each end. A /30 subnet gives exactly 2 usable addresses ($2^2 - 2 = 2$) and wastes the least space.

Link	Network Address	Subnet Mask	Router A (DCE end)	Router B	Broadcast
R1 — R2	192.168.1.152	255.255.255.252 (/30)	R1 Se0/0 .153	R2 Se0/0 .154	.155
R2 — R3	192.168.1.156	255.255.255.252 (/30)	R2 Se0/1 .157	R3 Se0/0 .158	.159
R1 — R3	192.168.1.160	255.255.255.252 (/30)	R1 Se0/1 .161	R3 Se0/1 .162	.163

After allocating all 8 subnets, addresses **192.168.1.164 to 192.168.1.255** remain unused (92 addresses). These are available if the company adds more departments later.

3.4 Complete Subnet Summary

The table below brings everything together in one place for easy reference.

#	Department / Link	Network	Mask	First Host	Last Host	Broadcast	Usable
1	IT	192.168.1.0	/26	.1	.62	.63	62
2	HR	192.168.1.64	/27	.65	.94	.95	30
3	Finance	192.168.1.96	/27	.97	.126	.127	30
4	Sales	192.168.1.128	/28	.129	.142	.143	14
5	Management	192.168.1.144	/29	.145	.150	.151	6
6	WAN R1–R2	192.168.1.152	/30	.153	.154	.155	2
7	WAN R2–R3	192.168.1.156	/30	.157	.158	.159	2
8	WAN R1–R3	192.168.1.160	/30	.161	.162	.163	2

4. Router Interface IP Addresses

Each router interface is given the first usable address of the subnet it serves. The table below shows every interface, what it connects to, and its IP address.

Router R1

Interface	Connects to	IP Address	Subnet Mask
Fa0/0	SW-IT (IT Dept)	192.168.1.1	255.255.255.192
Fa0/1	SW-HR (HR Dept)	192.168.1.65	255.255.255.224
Se0/0	R2 — WAN link	192.168.1.153	255.255.255.252
Se0/1	R3 — WAN link	192.168.1.161	255.255.255.252

Router R2

Interface	Connects to	IP Address	Subnet Mask
Fa0/0	SW-Finance (Finance Dept)	192.168.1.97	255.255.255.224
Fa0/1	SW-Sales (Sales Dept)	192.168.1.129	255.255.255.240
Se0/0	R1 — WAN link	192.168.1.154	255.255.255.252
Se0/1	R3 — WAN link	192.168.1.157	255.255.255.252

Router R3

Interface	Connects to	IP Address	Subnet Mask
Fa0/0	SW-Mgmt (Management Dept)	192.168.1.145	255.255.255.248
Se0/0	R2 — WAN link	192.168.1.158	255.255.255.252
Se0/1	R1 — WAN link	192.168.1.162	255.255.255.252

5. End Device IP Addresses

Each PC is configured with a static IP address from its department's subnet. The default gateway is the router interface connected to that department.

PC	Department	IP Address	Subnet Mask	Default Gateway
PC-IT1	IT	192.168.1.2	255.255.255.192	192.168.1.1
PC-IT2	IT	192.168.1.3	255.255.255.192	192.168.1.1
PC-HR1	HR	192.168.1.66	255.255.255.224	192.168.1.65
PC-HR2	HR	192.168.1.67	255.255.255.224	192.168.1.65
PC-F1	Finance	192.168.1.98	255.255.255.224	192.168.1.97
PC-F2	Finance	192.168.1.99	255.255.255.224	192.168.1.97
PC-S1	Sales	192.168.1.130	255.255.255.240	192.168.1.129
PC-S2	Sales	192.168.1.131	255.255.255.240	192.168.1.129
PC-M1	Management	192.168.1.146	255.255.255.248	192.168.1.145
PC-M2	Management	192.168.1.147	255.255.255.248	192.168.1.145

6. Routing Configuration

6.1 Routing Plan

The assignment requires two static routes and dynamic routing for the rest. I chose **RIP version 2** as the dynamic protocol because it is simple to configure and supports VLSM with the **no auto-summary** command.

The two static routes I configured are:

1. R1 has a static route to the Management subnet (192.168.1.144/29) via R3's serial interface (.162). This simulates a fixed management path that the admin controls directly.
2. R3 has a static route to the IT subnet (192.168.1.0/26) via R1's serial interface (.161). This demonstrates a second static route going in the other direction.

All other subnets are learned automatically through RIP v2.

6.2 R1 Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname R1

! --- LAN interfaces ---
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.192
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface FastEthernet0/1
R1(config-if)# ip address 192.168.1.65 255.255.255.224
R1(config-if)# no shutdown
R1(config-if)# exit

! --- WAN serial interfaces (R1 is DCE, so clock rate goes here) ---
R1(config)# interface Serial0/0
R1(config-if)# ip address 192.168.1.153 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface Serial0/1
R1(config-if)# ip address 192.168.1.161 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit

! --- Static route 1: Management via R3 ---
R1(config)# ip route 192.168.1.144 255.255.255.248 192.168.1.162
```

```
! --- RIP version 2 ---
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# no auto-summary
R1(config-router)# network 192.168.1.0
R1(config-router)# network 192.168.1.64
R1(config-router)# network 192.168.1.152
R1(config-router)# network 192.168.1.160
R1(config-router)# exit
R1# write memory
```

6.3 R2 Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname R2

R2(config)# interface FastEthernet0/0
R2(config-if)# ip address 192.168.1.97 255.255.255.224
R2(config-if)# no shutdown
R2(config-if)# exit

R2(config)# interface FastEthernet0/1
R2(config-if)# ip address 192.168.1.129 255.255.255.240
R2(config-if)# no shutdown
R2(config-if)# exit

R2(config)# interface Serial0/0
R2(config-if)# ip address 192.168.1.154 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit

! --- R2 is DCE on the R2-R3 link ---
R2(config)# interface Serial0/1
R2(config-if)# ip address 192.168.1.157 255.255.255.252
R2(config-if)# clock rate 64000
R2(config-if)# no shutdown
R2(config-if)# exit

R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 192.168.1.96
R2(config-router)# network 192.168.1.128
R2(config-router)# network 192.168.1.152
R2(config-router)# network 192.168.1.156
R2(config-router)# exit
R2# write memory
```

6.4 R3 Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname R3

R3(config)# interface FastEthernet0/0
R3(config-if)# ip address 192.168.1.145 255.255.255.248
R3(config-if)# no shutdown
R3(config-if)# exit

R3(config)# interface Serial0/0
R3(config-if)# ip address 192.168.1.158 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# exit

R3(config)# interface Serial0/1
R3(config-if)# ip address 192.168.1.162 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# exit

! --- Static route 2: IT Dept via R1 ---
R3(config)# ip route 192.168.1.0 255.255.255.192 192.168.1.161

R3(config)# router rip
R3(config-router)# version 2
R3(config-router)# no auto-summary
R3(config-router)# network 192.168.1.144
R3(config-router)# network 192.168.1.156
R3(config-router)# network 192.168.1.160
R3(config-router)# exit
R3# write memory
```

7. Testing — Ping Verification

After all configurations were entered and saved I tested connectivity from different PCs across the network. The first ping sometimes times out while RIP is still converging; waiting about 30 seconds and trying again gives a successful response.

The commands below show how to run the tests from the Packet Tracer CLI on any PC:

```
! From PC-IT1 — test reach to all other departments
PC> ping 192.168.1.66       (HR  dept — PC-HR1)
PC> ping 192.168.1.98       (Finance — PC-F1)
PC> ping 192.168.1.130      (Sales   — PC-S1)
PC> ping 192.168.1.146      (Mgmt    — PC-M1)

! Verify routing table on R1
R1# show ip route
R1# show ip protocols
R1# show ip interface brief
```

To confirm the static routes are present:

```
R1# show ip route static
R3# show ip route static
```

All pings returned successful replies once RIP had converged. The *show ip route* output on each router showed the correct routes learned via RIP (marked R) and the two manually entered routes (marked S).

8. Conclusion

In this assignment I designed a five-department company network using VLSM to make the most efficient use of the 192.168.1.0/24 address space. By allocating subnets from largest to smallest I was able to fit all five departments plus three WAN links inside the /24 block while keeping 92 addresses free for future growth.

The two static routes give the administrator direct control over specific paths in the network, while RIP v2 handles the rest of the routing automatically. All PCs across all five departments can successfully ping each other, confirming that the design works correctly.